CT Scan Abnormality Detection System

Engineering of AI-Intensive Systems

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1. Goal of the System

The main purpose of the system is to assist radiologists in identifying abnormalities in a patient's CT scan based on a trained dataset. By adding the system to the pipeline for identifying abnormalities on a CT scan, it will accelerate the detection process. This provides a high-confidence second opinion, significantly reducing diagnostic delays and the required time to create and finalize the CT scan report, especially in emergencies.

1.1 Non-AI-Related Goals

- Assist radiologists in efficiently analyzing CT scans.
- Reduce diagnostic delays and clinical bottlenecks, especially within high-pressure emergency department contexts.
- Provide a secondary opinion to increase general diagnostic confidence.
- Reduce overall patient radiation dose exposure over time.

1.2 AI-Related Goals

- Automatically detect localized abnormalities within CT scan data using computer vision.
- Highlight suspicious regions of interest within CT images to support rapid manual review.
- Enhance the underlying image quality, clarity, and contrast of the scan through deep learning preprocessing models.

2. Stakeholders

- **Users:** Radiologists who interact directly with the application to review scans and write clinical documentation.
- **People Affected by the System:** Patients whose diagnostic timelines, physical safety, and treatment plans depend on the precision of the software.
- **Managers:** Professors, medical researchers, and hospital administrators who track system output and deployment metrics.
- **Regulators:** Healthcare regulatory authorities enforcing data protection acts, medical software certifications, and patient safety compliance.

3. System Requirements

System requirements are defined below using the EARS framework.

3.1 Functional Requirements (FR)

- **FReq:** When a CT scan is uploaded, the CT application shall detect and identify potential abnormalities. *(AI-Specific)*
- **FReq:** When a CT scan is uploaded, the CT application shall compare the scan with stored data to identify similar abnormalities. *(AI-Specific)*

- **FReq:** When a CT scan is uploaded, the CT application shall retrieve and present similar cases from the dataset. *(AI-Specific)*
- **FReq:** When a CT scan is uploaded, the CT application shall provide a probability percentage for each possible abnormality diagnosis. *(AI-Specific)*

3.2 Non-Functional Requirements (NFR)

- **NfReq:** The CT application shall allow users to authenticate with an account.
- **NfReq:** When the user authenticates, the CT application shall display the main home screen dashboard viewer.
- **NfReq:** When the user presses search, the CT application shall find a patient based on their unique Patient ID.
- **NfReq:** When the user is uploading a scan, the CT application shall process the CT scan matrix and provide the output result.
- **NfReq:** The response time of the system for model inference should be fast, around 30 seconds or less. *(AI-Specific)*
- **NfReq:** Since it is in the medical field, the analytical diagnostic accuracy of the model should be at least 90%. *(AI-Specific)*
- **NfReq:** The CT application shall be updated once a month with the new patient results, and it should be done at night to avoid clinical downtime. *(AI-Specific)*
- **NfReq:** Only authorized personnel shall access the application.
- **NfReq:** All CT scans and patient history data shall be encrypted at rest and in transit.
- **NfReq:** The CT application shall be trained on a wider range of demographics to avoid dataset and demographic bias. *(AI-Specific)*

4. Use Case Descriptions

4.1 Main Use Cases

Use Case 1: Login to CT Application (UC1)

- **ID:** UC1
- **Description:** Authenticates the radiologist and grants secure workspace access.
- **Actors:** Radiologist
- **Stakeholders:** Patient, Radiologist, Hospital Administration (Managers)
- **Preconditions:** The platform network is online, and the radiologist holds a pre-registered account.
- **Success End Condition:** Successful authentication and redirection to the application dashboard.
- **Failure End Condition:** Access denied; user credentials remain unverified.
- **Main Success Scenario:**
    - Radiologist navigates directly to the portal login interface.
    - System displays the interactive credentials login form.
    - Radiologist inputs their specific username and password, then submits.
    - System securely validates the incoming credentials against the authorization layer (includes **SUC1**).
    - System creates a secure authenticated session token state.

  - System redirects the radiologist safely to the main functional dashboard.
- **Alternative Scenarios:**
  - *A1:* Radiologist enters invalid credentials → The system displays an error message and prompts a retry (Go back to step 3).
- **Exception Scenario:**
  - *E1:* Security mechanism intercepts 3 successive failed authentication attempts → The account is locked out for safety; the system alerts the admin.

Use Case 2: Find Patient (UC2)

- **ID:** UC2
- **Description:** Locates an individual patient file using unique identifiers.
- **Actors:** Radiologist
- **Stakeholders:** Patient, Radiologist, Manager
- **Pre-Conditions:** Radiologist is actively authenticated; target patient profile exists within database structures.
- **Success End Condition:** Target patient record successfully retrieved and rendered on screen.
- **Failure End Condition:** Patient profile cannot be identified or located.
- **Main Success Scenario:**
  - Radiologist clicks onto the "Search" tool action icon on the dashboard.
  - System displays the query search interface panels.
  - The radiologist enters the target unique patient ID parameter.
  - System queries database indexers looking for structural matches.
  - System renders a dynamic list of matching patient entries.
  - Radiologist clicks and selects the exact target patient record.

- **Alternative Scenarios:**
  - *A1:* Patient ID isn't directly known → Radiologist queries using combinations of patient name and date of birth.
- **Exception Scenario:**
  - *E1:* Query yields zero results → System throws "ID not found" warning and prompts user to instantiate a new profile.

Use Case 3: Upload CT Scan (UC3)

- **ID:** UC3
- **Description:** Uploads raw imaging files safely onto the diagnostic platform.
- **Actors:** Radiologist
- **Stakeholders:** Patient, Radiologist, Hospital Managers
- **Preconditions:** Radiologist is fully authenticated, and a patient profile is actively selected.
- **Success End Condition:** The scan is stored in secure architecture and mapped to the patient profile.
- **Failure End Condition:** Network drop or file invalidation terminates the upload sequence.
- **Main Success Scenario:**

- ○ Radiologist selects the "Upload Scan" function item on the dashboard view.
  - ○ System displays file exploration dialogue boxes.
  - ○ Radiologist selects a target local image file and confirms transmission.
  - ○ System validates file format compliance and structure integrity.
  - ○ System processes data through the pipeline encryption engine (includes **SUC2**).
  - ○ System saves files into secure storage structures linked to the Patient ID.
  - ○ The system alerts the radiologist that the file upload succeeded.
- **Alternative Scenarios:**
  - *A1:* File type uploaded is DICOM standard → System blocks upload, outputs "Invalid format," and defaults back to step 3.
- **Exception Scenario:**
  - *E1:* The transport layer drops the connection during the file stream → Uploads abort, partial buffers are purged, and error codes are flagged.

Use Case 4: Analyze CT Scan (UC4)

- **ID:** UC4
- **Description:** Employs computer vision inference models to segment abnormalities and process diagnostics.
- **Actors:** Radiologist, Image Processor AI Model
- **Stakeholders:** Patient, Radiologist, Manager
- **Preconditions:** Target CT imaging file is successfully parsed and present on storage systems.
- **Success End Condition:** Imaging features are structurally extracted, classified, and localized accurately.
- **Failure End Condition:** Algorithmic breakdown or timeout prevents accurate model generation.
- **Main Success Scenario:**
  - System fetches raw voxel matrices from secure storage objects (includes **SUC3**).
  - The deep learning image processor model computes feature maps over the CT scan.
  - System calculates conditional statistical probability vectors for features (includes **SUC5**).
  - System runs context matching vectors across historical patient indices (includes **SUC4**).
  - System writes out all preliminary model inferences back into relational database stores.
- **Alternative Scenarios:**
  - *A1:* Input scan exhibits image noise → The system triggers automated image pre-processing filters to boost quality before inference.
- **Exception Scenario:**
  - *E1:* Model inference engine processing bottlenecks exceed the 30-second boundary threshold → The system triggers a warning alert to the workspace interface and requests the operator reset the job.

Use Case 5: Review Diagnosis (UC5)

- **ID:** UC5
- **Description:** Radiologist audits AI predictions to compile a validated clinical diagnostic report.
- **Actors:** Radiologist
- **Stakeholders:** Patient, Radiologist, Medical Researchers
- **Preconditions:** Radiologist is authenticated, and target backend machine learning analysis is completed.
- **Success End Condition:** Final clinical diagnosis is saved and authenticated with electronic signatures.
- **Failure End Condition:** Database saves the fault, preventing record modification.
- **Main Success Scenario:**
    - Radiologist selects and opens the newly processed scan visualization.
    - System renders the visual CT slice layer augmented with highlighted regions and probability score metrics (**Includes SUC5**).
    - Radiologist evaluates AI suggestions alongside contextually provided historical similar case profiles (**Includes SUC4**).
    - Radiologist records definitive clinical conclusions and submits a digital sign-off signature.
    - System permanently updates core EHR records with the finalized report status.
- **Alternative Scenarios:**
    - *A1:* Radiologist diagnostic perspective differs from AI recommendations → Operator overrides system suggestions and logs customized manual parameters.
- **Exception Scenario:**
    - *E1:* Critical transactional fault blocks EHR database write-outs during file save → The system preserves data within the local application runtime cache and flags alerts to infrastructure admins.

4.2 Supporting Use Cases (SUC)

Supporting Use Case 1: Authenticate Credentials (SUC1)

- **ID:** SUC1
- **Description:** Validates raw authentication data using secure cryptographic matches.
- **Actors:** Radiologist
- **Pre-Conditions:** Secure identity service endpoints are operational and accessible.
- **Success End Condition:** Safe verification confirmed; session verification generated.
- **Main Success Scenario:**
    - Security layer receives plaintext password strings from the view forms.
    - System pulls target identity hashes corresponding to the username from storage tables.
    - System runs secure cryptographic hashing calculations to cross-verify keys.

- System routes an explicit "Success" or "Failure" status back into the parent login use case flow.
- **Exception Scenario:**
  - *E1:* Central identity databases are un-routable → System logs service interruption and handles a "Service Unavailable" error state.

Supporting Use Case 2: Encrypt Patient Data (SUC2)

- **ID:** SUC2
- **Description:** Applies corporate-grade encryption routines over electronic files.
- **Actors:** System
- **Pre-Conditions:** Target data structures are actively cached within memory buffers.
- **Success End Condition:** Files are committed into physical storage blocks exclusively in encrypted formats.
- **Main Success Scenario:**
  - System ingests raw patient structured data records.
  - The encryption engine executes cipher algorithms using secret key pairs.
  - System checks block integrity hashes to ensure no transmission corruption took place.
  - System routes a completed "Done" code token to parent handler.
- **Exception Scenario:**
  - *E1:* Software encryption engines fault out mid-cycle → The system flushes raw data leaks out of memory buffers instantly and issues emergency admin notifications.

Supporting Use Case 3: Detect Abnormalities (SUC3)

- **ID:** SUC3
- **Description:** Scans images to detect structural trends and assign spatial coordinates over visual bounds.
- **Actors:** AI Inference Model
- **Pre-Conditions:** Image matrices are fully decompressed and mapped to RAM.
- **Success End Condition:** Localized coordinate bounding regions are identified and formatted.
- **Main Success Scenario:**
  - System processes image tensors through the neural network layers.
  - Architecture maps mathematical activations to known abnormal pixel structural patterns.
  - System calculates precise dimensional coordinate maps outlining spatial bounds.
  - System compiles coordinate datasets into a visual vector graphics layer routed to the frontend view.
- **Exception Scenarios:**
  - *E1:* Model resolves zero structural deviations → The system outputs an empty findings array without throwing operational failure flags.

    - *E2:* Spatial noise ratios breach threshold tolerance metrics → The system logs an inability to parse cleanly and triggers image enhancement fallbacks.

Supporting Use Case 4: Retrieve Similar Cases (SUC4)

- **ID:** SUC4
- **Description:** Queries global databases for historical scans matching current visual vector signatures.
- **Actors:** AI Inference Model
- **Pre-Conditions:** Abnormality pattern vectors have been generated by upstream neural layers.
- **Success End Condition:** Historical clinical context objects are fetched and structured into workspace memory.
- **Main Success Scenario:**
    - System matches incoming anomaly vectors across indexed reference datasets.
    - System extracts metadata summaries from identical historical case files.
    - System computes context confidence metrics evaluating clinical file matches.
    - System bridges data payloads back into core application threads.
- **Exception Scenarios:**
    - *E1:* Vector lookup database requests time out→ System catches the fault, defaults to an empty payload object, and safely logs the timeout event.
    - *E2:* Historical archives are locked out for scheduled data upgrades → The request fails, and application interfaces hide reference panels temporarily.

Supporting Use Case 5: Calculate Probability Percentage (SUC5)

- **ID:** SUC5
- **Description:** Computes rigorous statistical confidence scores for detected classifications.
- **Actors:** AI Inference Model
- **Pre-Conditions:** Neural feature maps have isolated candidate structures.
- **Success End Condition:** Softmax/confidence vectors are parsed into user-friendly diagnostic percentages.
- **Main Success Scenario:**
    - System evaluates activation intensities against baseline validation model weights.
    - Engine calculates bounded statistical confidence intervals.
    - System translates raw floating-point probabilities into user-facing percentages.
    - System binds formatting metrics straight onto diagnostic reporting blocks.
- **Exception Scenario:**
    - *E1:* Highly contradictory feature distributions create an indeterminate mathematical state → System aborts estimation calculation, records an "Analysis Error," and flags a manual clinician review token.

5. Traceability Matrix

Use cases	UC1	UC2	UC3	UC4	UC5	SUC1	SUC2	SUC3	SUC4	SUC5
Requirements										
FR1	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
FR2	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
FR3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
FR4	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	TRUE
NFR1	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE
NFR2	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NFR3	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NFR4	FALSE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE
NFR5	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NFR6	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
NFR7	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NFR8	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
NFR9	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
NFR10	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Fig 01. Requirement Traceability Matrix mapping all Functional Requirements (FR) and Non-Functional Requirements (NFR) to Main and Supporting Use Cases (UC/SUC) to ensure full validation and system verification.

The traceability matrix provides proof of fulfilling the requirements in main and supporting use cases by cross-referencing the requirements, such as the 30-second threshold and the 90% accuracy target. By linking the main requirements directly to the definitions in UC4 and UC5, it ensures that no goal or safety constraint remains unverified during the system implementation phase.

6. Use Case Diagram

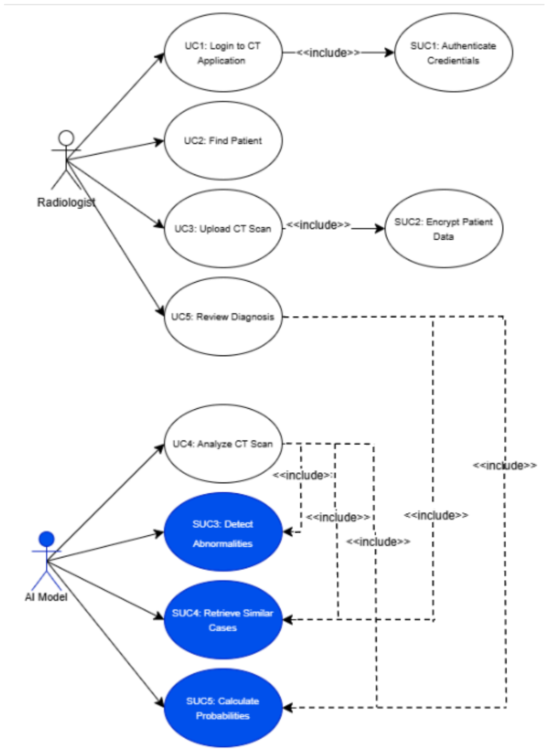

Fig 02. Use Case Diagram illustrating the behavioral boundaries of the system and the interaction patterns between human operators and the artificial intelligence model.

The model isolates two primary actors that are interacting with the system's boundaries: the radiologist, whose task is to drive the operations in use case 1 and use case 5, and the AI model, which acts as an automated computing agent during the abnormality detection. The separation of standard application features from the deep learning-specific routines is clearly visible in Fig. 02, showing how complex tasks such as retrieving similar scans from the historical dataset implicitly include some underlying supporting workflows.

7. Domain Model

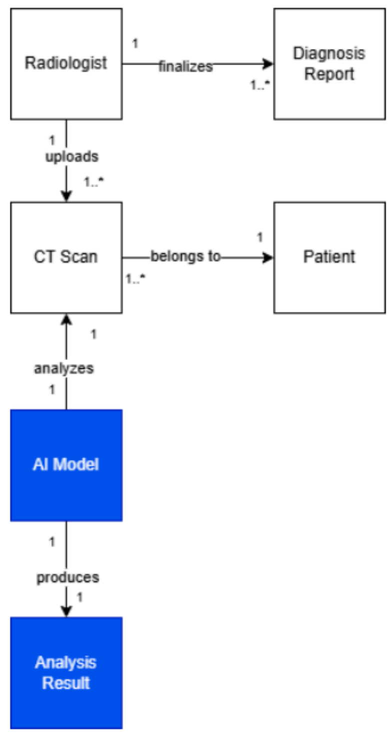

Fig 03. Conceptual Domain Model capturing core data entities, relationship multiplicity, and AI-specific domain extensions engineered for patient safety.

The domain model establishes the data structure that is required to support both clinical operations and machine learning tasks while preventing unauthorized data exposure. Standard medical components, including encrypted patient profiles and image metadata records, are mapped with structural boundaries to distinguish them from AI-specific domain extensions. These extensions explicitly model unstable runtime variables, such as spatial coordinate arrays.

8. System Architecture & Component Descriptions

8.1 Architecture Diagram

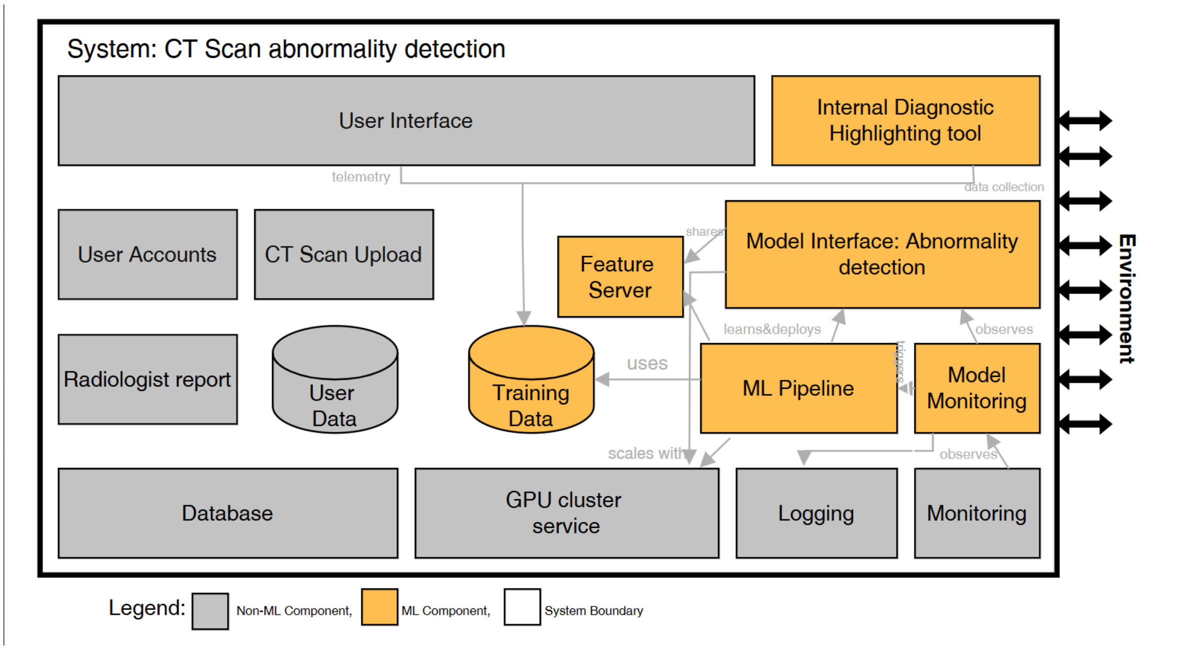

Fig 04. Microservice System Architecture Diagram displaying decoupled data paths, secure network boundaries, and accelerated compute infrastructures.

The system architecture uses a scalable, decoupled microservice pattern, designed to prevent clinical downtime and to handle high data volumes through cloud-based infrastructure. The diagram shows the execution path of a single scan, from the web-based frontend (the user interface) to the asynchronous file ingestion layer. It shows the strict logical isolation between the core application database and the machine learning pipeline components and how the Feature Server and Model Interface API use the GPU Cluster Service to provide low-latency responses while shielding day-to-day hospital operations.

8.2 Component Layout

Non-ML Components:

- User Interface: This is the web-based portal where the radiologist interacts with the system. It is linked with requirements NfReq NFR1/2/3/4, and it handles the login, display, search, and upload triggers.
- User Accounts: This is where the accounts are stored and authenticated during login. Linked with NfReq < NFR1>.
- Radiologist Report: This component has the creation, storage, and finalization of the report. It has the analysis results approved by the radiologist.
- User Data: A database containing patient metadata such as name, age, and medical history, which is different from the database.
- CT Scan Upload: The component that handles the files, performs initial data integrity, and triggers the ML pipeline. The linked requirement is NfReq <NFR4>.
- Database: Where the user IDs, hashed passwords, and files are stored. It connects to the backend and user accounts. Linked requirements are NfReq <NFR9>.
- GPU Cluster Service: This is needed for faster and secure training. Linked with requirement NfReq <NFR5>
- Logging: This component registers the record of system activities, including who accessed what and when. This is linked with requirement NfReq <NFR8>, only authorized access is allowed.
- Monitoring: This is tracking the system in order to ensure its health. It is making sure the application doesn't crash. This could be linked to requirement NfReq <NFR5>, where we ensure the results are generated in less than 30 seconds.

ML Components:

- Internal Diagnostic Highlighting Tool: This component is the visual overlay. It takes the coordinates provided by the model interface and renders the bounding boxes directly onto the CT scan. The linked requirements are Req <FR1>.
- Feature Server: This is where the embeddings of historical scans are stored. When the radiologist uploads a new scan, the system compares its features against the database to find similar cases. Linked requirements are Req <FR2> and Req <FR3>.
- Training Data: The data used for training. It should be encrypted and unbiased, linking it to NfReq <NFR9> and <NFR10>.
- ML Pipeline: The pipeline manages the monthly retraining of the model. It uses the GPU cluster service to achieve high-performance compute, which is necessary to process the CT scan within the time limit. Linked requirements are NfReq <NFR5> for the time limit and NfReq <NFR7> for automating the update.
- Model Interface: Abnormality Detection: This hosts the model. It receives the CT data, runs inference, and returns coordinates and probability scores. It is linked with requirements Req <FR1> and <FR4>, as well as NfReq <NFR6> with accuracy.
- Model Monitoring: This tracks the AI's performance, making sure the accuracy stays high. It is linked with requirement NfReq <NFR6>.

8.3 Architectural Design Questions & Answers

1. What is the required level of accuracy?

The system must achieve an overall detection accuracy higher than 90%. Specifically, it must prioritize sensitivity (recall) to ensure no positive abnormalities are missed, as the clinical cost of a false negative is significantly higher than a false positive.

2. What constitutes an "unacceptable mistake," and how is it avoided?

The most critical mistake is automation bias, where a clinician relies entirely on the AI's output without verification. This is avoided by implementing a "Human-in-the-Loop" requirement, where the system serves as a decision-support tool that explicitly requires a radiologist's sign-off for every report.

3. How will the system capture feedback?

Feedback is captured through weekly reviews by professional radiologists. Additionally, the system uses a telemetry stream to monitor real-time user corrections in the dashboard, which are then fed back into the ML pipeline for future model retraining.

4. Is there a safety risk involved?

Yes, a false negative prediction poses a direct risk to patient life. To mitigate this, the system is engineered with clinical guardrails, and any high-discrepancy cases are flagged for immediate secondary review.

5. How will the system handle a failed analysis?

If the inference engine fails or produces an error, the details are captured by the centralized logging component and reported to the system monitoring dashboard. This triggers an immediate technical alert while the UI reverts to a "manual-only" viewing mode to ensure clinical continuity.

6. Can we offer this as an app or a website?

The service will be offered as a web service. This is optimal for hospital environments, as it allows radiologists to access high-resolution imaging data from any medical workstation without requiring local software installations.

7. Should the system learn from patient data?

Yes, the system should employ continuous learning to improve detection accuracy over time. However, all data must undergo a de-identification process to remove protected health information (PHI) before it is stored in the training & validation datasets.

8. How will the system handle a high volume of data per day?

The system utilizes cloud infrastructure to achieve scalability. Specifically, the GPU cluster service can dynamically expand to handle spikes in scan uploads, ensuring that performance remains stable regardless of the number of active users.

9. Can we estimate the quality of the AI model's predictions?

Yes, quality is estimated by comparing AI predictions against the ground truth provided by radiologists. The system also tracks prediction confidence scores for every scan to flag cases where the AI is uncertain.

10. Is there a chance of leaking patient data?

Risk is minimized through strict encryption and data isolation. Each patient is assigned an encrypted ID, and the system boundary is designed to ensure that identifiable personal data never reaches the machine learning training environment.